Kubernetes Production Deployment Standard Operating Procedure

1. Canary Deployment Verification Strategy

All production releases must follow a canary progressive delivery strategy using Argo Rollouts. Initial canary traffic weight must be set to 5% for 15 minutes while monitoring P99 latency and HTTP 5xx error rates.

2. Resource Limits & Horizontal Pod Autoscaling

Every production pod must declare CPU and Memory requests and limits. Horizontal Pod Autoscaler (HPA) must target 70% average CPU utilization with minimum 3 replicas and maximum 25 replicas across multi-AZ clusters.